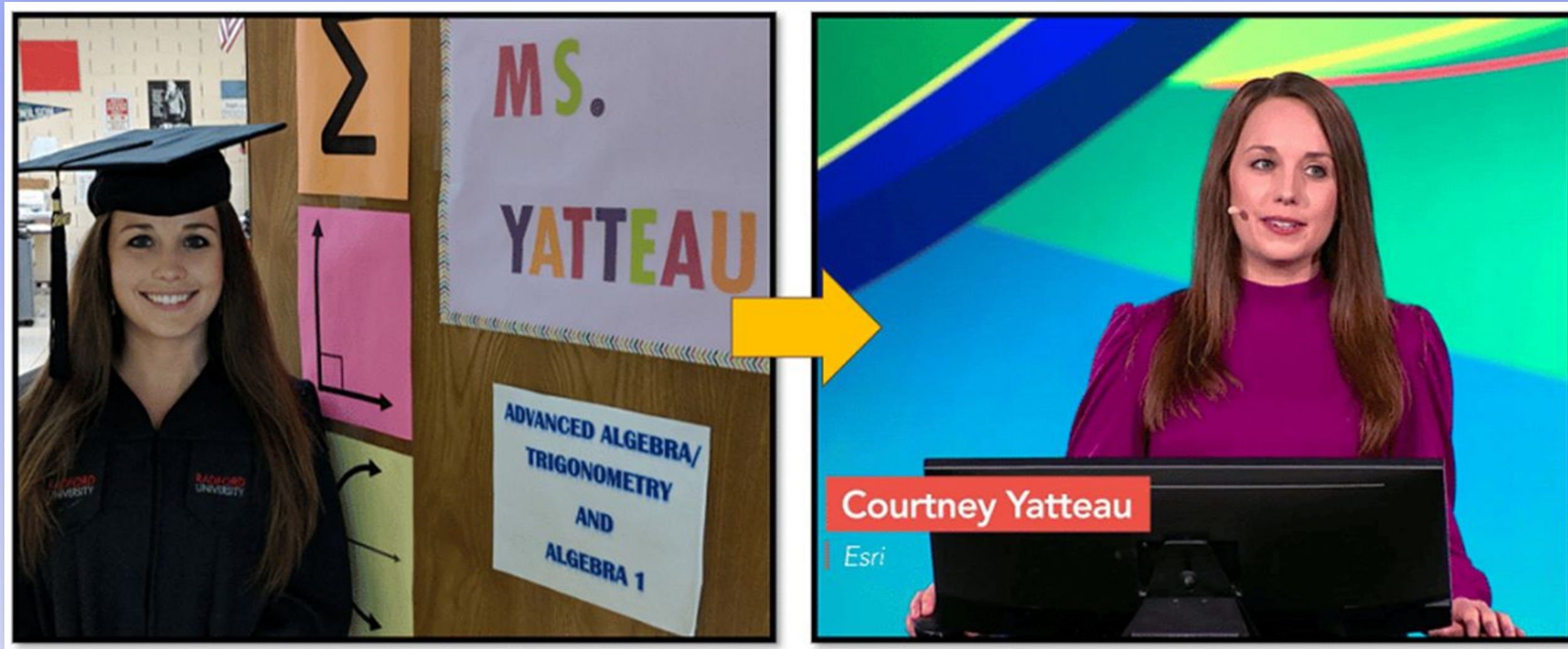




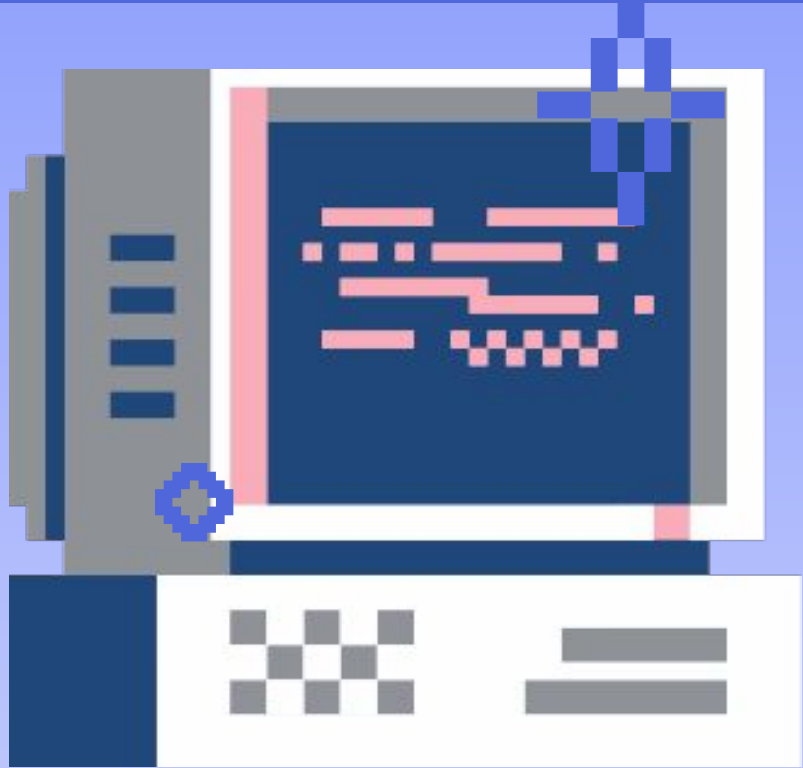
AI-Powered Gamification for the Web

Courtney Yatteau

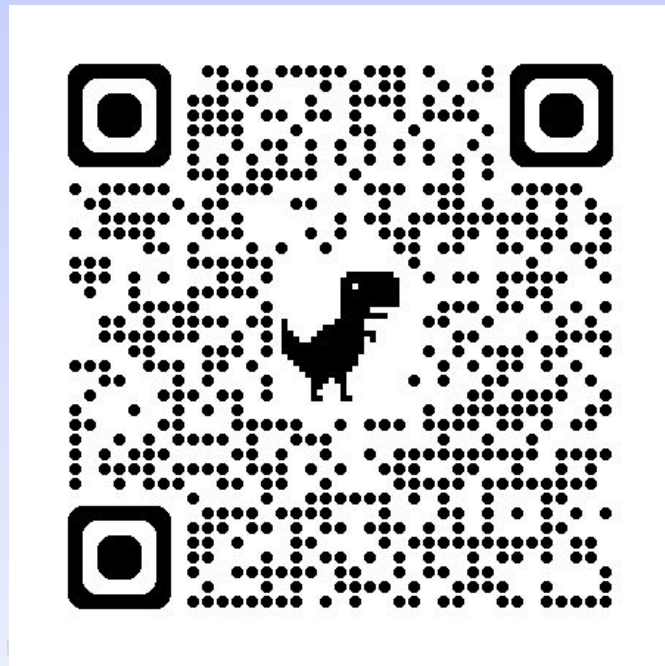


Courtney Yatteau

Developer Advocate, Esri



Demos and Slides



github.com/cyatteau/ndc-copenhagen-2026-ai-gamification

Agenda

- 1 Gamification basics
- 2 Three practical AI patterns
- 3 One combined app
- 4 How to build this in practice



Why Gamification Might Feel Shallow



Points feel meaningless

Badges no one cares about

Feedback feels generic

Difficulty feels random





PROGRESS



FEEDBACK

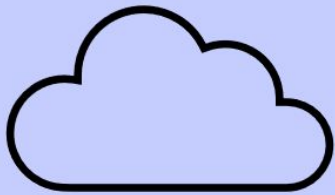


REWARD



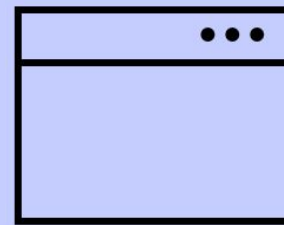
CHALLENGE

AI gamification is practical now



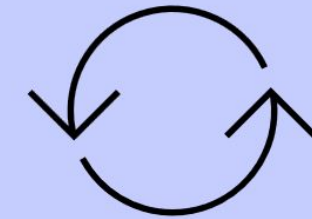
Hosted AI

Generate
Classify



Browser AI

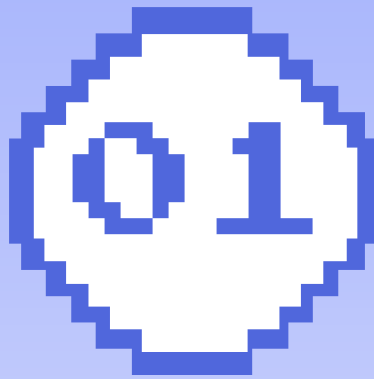
Adapt locally
Fast feedback



Smaller AI loops

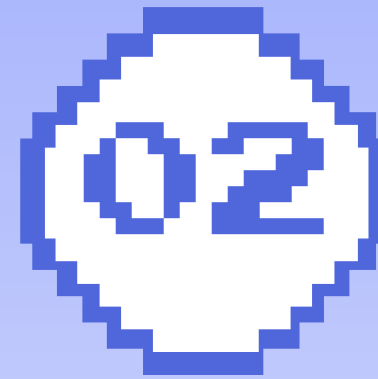
One useful signal
One useful change

Don't overbuild.



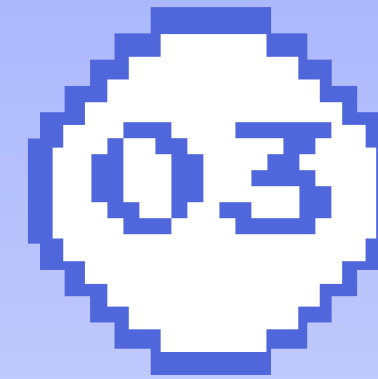
SIGNAL

What the app
observes



DECISION

What the AI does
with that signal



RESPONSE

How the UI
changes

Dynamic Content

Generate fresh prompts, facts,
or next-step variations
without a giant content system

VARIETY

REPLAYABILITY

LIGHTWEIGHT PERSONALIZATION



MINI DEMO 1 • DYNAMIC FACT REWARDS

Dynamic Fact Finder

Pick a place and a vibe. The app generates a short AI-powered fact, reveals a centered location snapshot after a couple of discoveries, and unlocks collectible postcards once the loop is rolling.

Choose a place

3441 N Fairfax Dr, Arlington, VA, 22201, USA

Pick the fact vibe

Fun

Spooky

Historical

Generate fact

Use my location

Reset

Fact generated. Map snapshot unlocked!

2 facts XP 20 Level 2 Map unlocked

Midnight Radar Echoes

Spooky Pin

Near Arlington’s busy corridor, 3441 N Fairfax Dr sits in a district where old roads, federal sites, and river fog have long fueled local ghost stories. After dark, the neighborhood feels a little too quiet for so much history.

PLACE

3441 N Fairfax Dr, Arlington, VA, 22201, USA

VIBE

Spooky

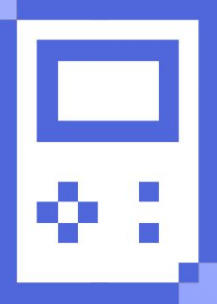
PROGRESS

Map snapshot unlocked. One more fact unlocks postcards.

Generate final unlock fact

Earned badges

First Find Map Reveal



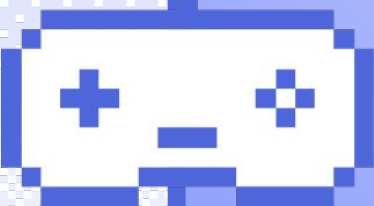
SIGNAL

Click
Fact type
Location



DECISION

Generate a structured fact card



RESPONSE

Fresh content
Reward label
Progress update

```
format: {  
  type: "json_schema",  
  strict: true,  
  schema: {  
    properties: {  
      headline: { type: "string" },  
      fact: { type: "string" },  
      rewardLabel: { type: "string" }  
    },  
    required: ["headline", "fact", "rewardLabel"]  
  }  
}
```

**Schema-shaped
output**

**Strict response
contract**

UI-friendly fields

What Powered this Demo



Place grounding

ArcGIS autosuggest
Place resolution
Reverse geocode



Content generation

OpenAI Responses API
Structured Outputs
GPT-5.4

Map reveal

Static Basemap Tiles
Centered snapshot crop
arcgis/community style

Reward

XP, badges, levels
GPT Image 1.5 postcard
Local unlock state

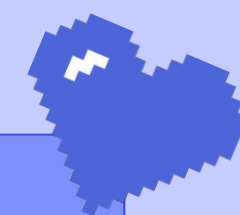


Server-side keys



HTTPS for location

Sentiment-Aware Feedback



React to tone and
choose a better feedback path

Tone Detection

Better Reinforcement

Smarter Follow-Up

Post-round checkpoint

MINI DEMO 2 • SENTIMENT ANALYSIS

Sentiment Snapshot

Share a quick reflection and get a tone-aware response that feels a little more thoughtful than the same generic message every time.

How did that round feel?

Example: I was frustrated at first, but I made it through.

Positive sample

Mixed sample

Frustrated sample

Clear

Analyze reflection

Sentiment

Waiting...

Response tone

Balanced

Top read

0%

Waiting

Your reflection will be summarized here.

Once you analyze a reflection, this area will give you a simple, user-facing read on the overall tone and the kind of response that fits it.

Tone mix

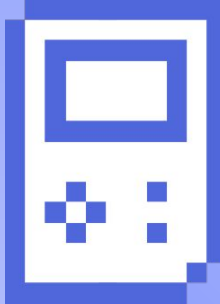
Positive

Neutral

Negative

Highlights

Analyze one reflection to see the standout lines.



SIGNAL

Player reflection

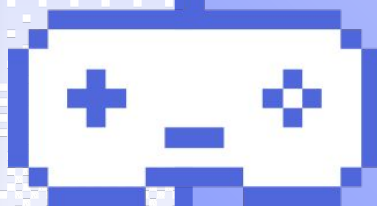
Sentence tone shifts



DECISION

Choose feedback route

Use confidence scores



RESPONSE

Tone changes

Recovery options change

```
const payload = {  
  kind: "SentimentAnalysis",  
  analysisInput: {  
    documents: [{ id: "1", language: "en", text }]  
  }  
};
```

Analyze the
reflection

```
const route = chooseRoute(doc);  
  
if (route.key === "negative") {  
  // protect progress, unlock support  
} else if (route.key === "mixed") {  
  // keep checkpoint, offer retry  
}
```

Choose a route

Change the UI



Player Input

Reflection after a round



Signal result

Document + sentence sentiment
Confidence scores

App response

Choose checkpoint route
Tone, support, and recovery

Adaptive Challenge Tuning

React to correctness, timing,
and repeated behavior
to keep challenge balanced

Behavior Signals Live Adjustment Better Pacing

MINI DEMO 3 · SMART DIFFICULTY

Adaptive Challenge

This quiz quietly adjusts its level based on how you are doing, so each round feels more responsive and less fixed. It is meant to feel polished and playable, not like a developer dashboard.

Status ● Smart pacing ready

Current **Easy**

Up next **Waiting...**

Accuracy **0%**

Best streak **3**

Easy

Round 0 of 8

Start a run to see the challenge adjust in real time.

Answer each question and the quiz will quietly tune what comes next.

Smart pacing is on

The next question will shift based on how this run is going, so the experience feels more tailored.

Start run

Reset

Round snapshot

THIS ROUND

Waiting for your first answer.

WHAT COMES NEXT

No next-step update yet.

WHY IT SHIFTED

The quiz will adjust after each response.

Difficulty balance

Easy

0%

Medium

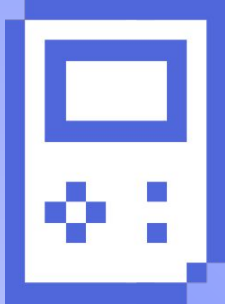
0%

Hard

0%

Run history

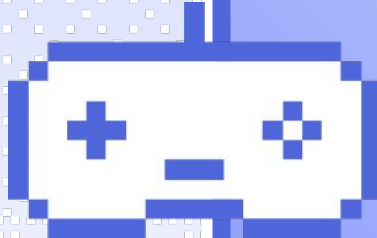
Start a run to build a quick history.



SIGNAL
Correctness
Response Time



DECISION
Next Prediction



RESPONSE
Current Difficulty

```
const features = [  
  correct ? 1 : 0,  
  responseTime,  
  currentDifficulty  
]
```

Capture features

```
const input = tf.tensor2d([features])  
const prediction = model.predict(input)  
currentDifficulty = prediction.argmax(-1).dataSync()[0]
```

Predict next state

```
difficultyLevel.textContent = ['Easy', 'Medium', 'Hard'][currentDifficulty]  
setTimeout(nextQuestion, 2000)
```

Update UI

What Powered this Demo



Browser-side model

TensorFlow.js

Small local prediction loop



Inputs

Correctness

Response time

Current difficulty

Decision + UI Response

Predict next difficulty

Change what comes next

Small model, noticeable pacing shift

The interesting part
isn't each feature.

It's how the loops
work together.



EXPLORE MORE

GeoExplorer

Tap around the map to uncover place-based challenges, save quick reactions, unlock new map styles, and build a living recap of your session.

Location challenges

Quick reactions

New map styles

Experience mode

Choose the full experience or focus on just one part of it.

Full experience

Challenges

Mood check-ins

Your progress

A simple view of how your session is going.

CURRENT PACE

Easy

ATTEMPTS

0

ACCURACY

0%

AVG TIME

0.0s

CURRENT STREAK

0

MOOD TAGS

0

Challenge pace

The experience quietly adjusts how challenging each place question feels based on your pace and accuracy.

Easy

Medium

Hard

Fresh start. The next challenge is leaning Easy.

Find address or place

1053 N Nelson St, Arlington, Virginia, 22201

PLACE CHALLENGE

1053 N Nelson St, Arlington, Virginia, 22201

Use the map and nearby context to make your best guess.

Current level: Easy. Quick city read

Quick city read

Easy level

In which Virginia city is this address located?

A

Arlington

B

Alexandria

C

Falls Church

D

Fairfax

Esri Community Maps Contributors | DCGIS | Arlington County, VA | Fairfax County, VA | VGIN | © OpenStreetMap, Microsoft | Esri | TomTom | Garmin | SafeGraph | GeoTechnologies, Inc | NASA | NGA | USGS | METI/NASA | FEMA | EPA | NPS | US Census Bureau | US... Powered by Esri

adaptive quiz mode

sentiment recap

badges + unlocks

The reusable loop

01

Signals in

clicks, answers, timing, tone, or streaks

02

AI decides

generate, classify, summarize, or adjust

03

UI responds

prompt, reward, unlock, recap or retune



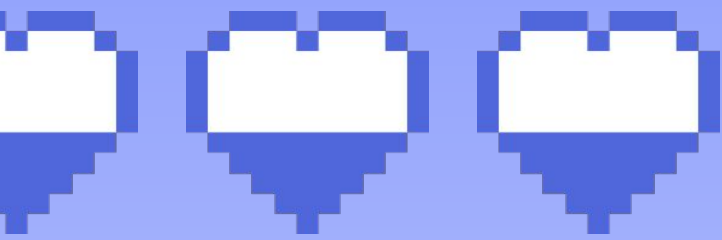
Better context

Better feedback loops

Visible UI change

Useful, not magical

THANKS FOR PLAYING



[c_yatteau](#)



[c_yatteau](#)



[cyatteau](#)

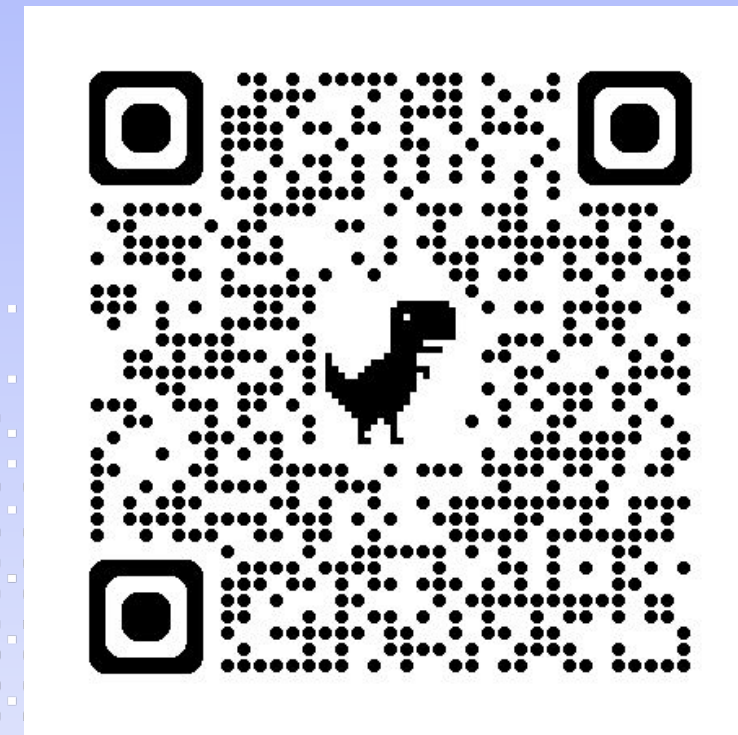


[courtneyyatteau](#)



[cyatteau.bsky.social](#)

Thank you, NDC Copenhagen!



github.com/cyatteau/ndc-copenhagen-2026-ai-gamification

GAME OVER